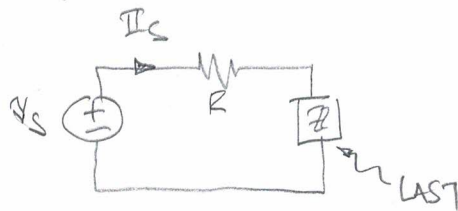


7.17)



$$\begin{cases} Z = 10 + j3 \Omega \\ V_s = 230 \angle 0^\circ \text{ V RMS} \\ R = 1 \Omega \end{cases}$$

a) EFFEKTEN (MESEN) SOM LEVERERES TIL LASTEN

Lösning:

$$\$_{\text{LAST}} = V_{\text{LAST}} I_s^* = Z I_s I_s^* = (10 + j3) |I_s|^2 \quad (1)$$

BEHÄNDER II.

$$I_s = \frac{V_s}{R + Z} = \frac{230 \angle 0^\circ}{1 + 10 + j3} = \frac{230 \angle 0^\circ}{11 + j3}$$

$$= \frac{230}{\sqrt{130}} \angle -15.25^\circ \text{ A RMS} \quad (2)$$

(2) i (1) ger

$$\$_{\text{LAST}} = (10 + j3) \left(\frac{230}{\sqrt{130}} \right)^2 = 406.92 (10 + j3)$$

MEDELEFFEKTEN RÖR

$$P = \text{Re}(\$_{\text{LAST}}) = \underline{4069.2 \text{ W}}$$

b) MEDELEFFEKT ABSORBERAT AV LINJEN

$$\$_{\text{LINJE}} = Z_{\text{LINJE}} |I_s|^2 = 1 \cdot \left(\frac{230}{\sqrt{130}} \right)^2 = 406.92 \text{ W}$$

↑
PÅRA-RÖR
EFFEKT.

c) KOMPLEX EFFEKT FÖR KLIVAN

$$\$ = V_s I_s = 230 \angle 0^\circ \frac{230}{\sqrt{130}} \angle -15.25^\circ = 4639.6 \angle -15.25^\circ$$

$$|\$| = \underline{4639.6 \text{ VA}}$$

d) LASTENS EFFEKTFAKTOR

$$\phi = \arctan\left(\frac{3}{10}\right) = \underline{16.7^\circ}$$

$$P_f = \cos \phi = \underline{0.958}$$

e) LINJE + LAST EFFEKTFAKTOR

$$P_f = \cos \phi = \cos(-15.25^\circ) \approx \underline{0.965}$$

SVAR:
se ovan